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Research Article

CNFI-NMI: A Robust Feature Selection Framework Integrating Complex Network Interactions and Normalized Mutual Information

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Abstract Feature selection plays an important role in understanding complex datasets across many disciplines, including finance, healthcare, and transportation. Conventional frameworks such as PGI-SDMI tend to be unstable and systemically biased because they depend on primitive proxies that lack coherence with model outputs and are sensitive to feature-scale heterogeneity. In this work, we introduce a novel framework, CNFI-NMI, that consists of two modules: 1) Complex Network Feature Interaction (CNFI), which incorporates higher-order associations among multiple metrics such as lift and support to robustly model interactions; and 2) Normalized Mutual Information (NMI), which enables unbiased relevance measurement independent of the scales of values. Experimental evaluations on three heterogeneous datasets covering traffic accidents, medical drug reactions, and financial risks indicate that CNFI-NMI consistently outperforms PGI-SDMI, especially on highly imbalanced classification problems. Our results show that accurate interaction-weight estimation drives performance gains more than the choice of graph-ranking algorithm. This framework combines a pragmatic balance of predictive power, generalizability, and interpretability.

Keywords Feature selection; Complex networks; Mutual information; Association rule mining; Feature interaction; Imbalanced data

1 Introduction

1.1 Background and Motivation

In the era of data-driven decision-making, machine learning has become an indispensable core technology across safety-critical fields. In financial risk assessment, models must navigate volatile market behaviors and fraudulent patterns to safeguard economic stability [1–3]. In precision medicine, identifying subtle biomarkers from complex genomic or clinical tabular data is crucial for early detection and treatment planning [4–7]. In transportation safety management, predicting accident risks and traffic flows from complex spatiotemporal data is key to optimizing urban mobility [8, 9]. Across all of these high-stakes scenarios, predictive performance and decision reliability are paramount.

In these domains, intricate dependencies create challenges that lie not merely in the number of raw features, but in the exponential growth of the interaction search space. Even with a moderate number of raw features, the potential high-order combinations constitute a "combinatorial explosion", effectively producing a "curse of dimensionality" within the interaction manifold [10]. This data characteristic severely restricts the efficiency, generalization ability, and ultimate trustworthiness of learning algorithms, particularly when dealing with tabular data structures [11, 12]. To confront this challenge, Feature Selection (FS)—identifying the most informative and predictive subset from the original feature set—remains a key preprocessing step [13, 14]. FS improves model performance, reduces computational overhead, and enhances interpretability.

An ideal feature selection method must address two core challenges: first, accurately quantifying the individual relevance of each feature to the target variable; and second, effectively capturing the complex interaction effects among features [13]. For example, in transportation safety prediction, "rainfall" or "evening rush hour" as individual features may indicate only a moderate risk, yet their combination ("rainfall during evening rush hour") can lead to an exponential increase

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in risk. Ignoring such interaction effects therefore incurs not only a loss in model performance but also an important cognitive bias regarding the true complexity of the system.

1.2 Related Work and Challenges

1.2.1 Taxonomy of Traditional Feature Selection Methods

Existing feature selection methods fall into three broad categories that embody distinct computational trade-offs: Filter, Wrapper, and Embedded methods [15]. Filter methods—such as those based on mutual information or statistical tests—are independent of the learning algorithm and thus provide important computational efficiency [16]. However, traditional filters tend to select redundant features because they ignore conditional dependencies between features [13, 17, 18]. Wrapper methods such as Recursive Feature Elimination (RFE) evaluate feature subsets through repeated model training, which yields strong performance advantages but incurs higher computational cost and overfitting risk. In other words, wrapper methods deliver better performance at high cost. Embedded methods—such as Lasso [19] and newer deep-learning-based approaches [20]—integrate feature selection into the model training process and thereby establish a critical balance between efficiency and performance. Nevertheless, this balance comes at a price: the selection outcome becomes tightly coupled with the specific learning algorithm [15], and this coupling constitutes a key limitation, since embedded methods inseparably link selection to the algorithm.

1.2.2 Evolution of Hybrid and Two-Stage Frameworks

Balancing efficiency with the capture of feature interactions has made hybrid and two-stage frameworks a significant research hotspot. Several studies have explored evolutionary and graph-based strategies to optimize this important trade-off. In particular, advanced swarm-intelligence algorithms have been adapted for multi-objective feature selection over complex tabular feature spaces [21–23], while fuzzy rough-set theories have been integrated with graph structures to model feature dependencies more accurately [24, 25]. Collectively, these frameworks address the efficiency–interaction trade-off effectively. A representative work among graph-based two-stage frameworks is the PGI-SDMI framework proposed by Feng et al. [26]. This framework first uses association rule mining [27] to construct a graph network that identifies interacting features, and then combines a customized mutual information metric so that individual features are evaluated more precisely. Although the method has shown potential in specific domains—proving the effectiveness of graph-network concepts in modeling feature interactions—key limitations still restrict its robustness, and these limitations, as the following analysis shows, reduce its overall reliability.

1.2.3 Technical Bottlenecks and Generalizability Challenges

The bottleneck of the interaction metric. When constructing the interaction network, current two-stage methods such as PGI-SDMI [26] rely solely on the "Confidence" metric [27] from association rules for edge weighting, which introduces significant limitations. Confidence is a fragile and incomplete metric, often biased by the marginal probability of the consequent [28]. Relying on a single metric is insufficient to capture complex non-linear interactions, a limitation also noted in recent graph-learning surveys [29], and the ablation experiments (Section 4.3) confirm that this single-metric reliance constitutes a primary performance bottleneck.

The fairness of the relevance metric.In its second stage, PGI-SDMI [26] employs a customized, non-normalized metric (SDMI), which compromises the fairness of the resulting scores. This introduces systemic bias when comparing features with different value ranges. By contrast, standardized metrics such as Normalized Mutual Information (NMI) [30] provide a fair comparison scale within the $[0, 1]$ interval. Because non-normalized metrics inject scale-dependent bias, they undermine reliable feature comparison, whereas NMI furnishes a fairer scale.

The challenge of generalizability. Owing to the limitations above, the effectiveness of such methods has been validated only within specific domains. Whether they can be transferred as a general framework to other domains with vastly different data characteristics remains uncertain. In particular, challenges such as high class imbalance leave generalizability an open question [15, 31]. Domain-specific validation alone is insufficient to support broad framework applicability, and existing methods therefore exhibit limited cross-domain generalizability.

1.3 Main Contributions

This paper proposes a general, robust feature selection framework named CNFI-NMI that systematically addresses the key limitations of existing two-stage methods. The framework is explicitly designed to balance performance sustainability, generalization robustness, and interpretability—three properties that are critical for prediction tasks over complex tabular data. We summarize the main contributions of the framework below, each of which

provides concrete support for the evidence presented in this paper, and together they show how the framework addresses the limitations identified above.

CNFI-NMI establishes a theoretically robust feature selection framework that organically fuses complex networks [32] with information theory [30]. By simultaneously addressing the two core limitations—the interaction-metric bottleneck and the relevance-metric unfairness—it achieves tangible generalizability across varied domains. To overcome the interaction-metric bottleneck, we design an innovative Composite Weight interaction module: a novel Composite Adaptive Weighting mechanism dynamically fuses multiple association metrics—such as support and lift [28]—to robustly quantify interaction strength, and is further combined with graph centrality [33, 34] for ranking. Through rigorous validation across three highly heterogeneous real-world datasets—transportation safety [35], medical diagnosis [36], and financial risk control [37]—we demonstrate the significant cross-domain validity of the framework. CNFI-NMI consistently and significantly outperforms its competitor PGI-SDMI [26] across the experiments presented in Section 4.2, and this advantage is especially critical on highly imbalanced financial data, where PGI-SDMI performs poorly. Beyond predictive accuracy, CNFI-NMI provides practical and interpretable structural insights through its feature interaction network, thereby supporting model-level explanation. Combined with downstream model-agnostic explanation techniques such as SHAP or LIME [38, 39], it offers both structural and behavioral interpretability. This dual capacity establishes CNFI-NMI as a relevant and significant contribution to the feature selection literature.

2 Problem Definition

This paper aims to construct a new, theoretically grounded, and high-performance feature selection paradigm by addressing the following three core challenges.

2.1 Challenge 1: Robust and Comprehensive Interaction

When constructing feature interaction networks, current methods (e.g., PGI-SDMI [26]) rely on a single association-rule metric (such as Confidence [40]), which is a one-sided simplification. Relationships between features in complex systems cannot be fully captured by simple conditional probabilities, because multi-dimensional metrics such as Lift and Support collectively define the value and strength of a rule [28]. Relying solely on one metric not only misses critical weak-signal interactions but can also be misled by accidental, spurious associations [13, 41, 42]. It is therefore essential to design a composite weighting mechanism capable of dynamically fusing multiple rule-evaluation metrics to more accurately quantify the non-linear dependency between features. In short, single-metric approaches miss weak signals and admit spurious associations.

2.2 Challenge 2: Theoretical Completeness and Fairness of Relevance Measurement

The effectiveness of a two-stage framework hinges on whether its different modules can be integrated on a unified and fair theoretical scale. If a framework (e.g., PGI-SDMI [26]) adopts a customized, unverified, and non-normalized metric in its second stage, it introduces systemic bias into feature comparison. This systemic bias in turn undermines reliable ranking across the framework's components. The second core challenge is therefore to adopt a metric with a solid theoretical foundation and standard normalization—such as Normalized Mutual Information (NMI) [30]—to guarantee an unbiased and reliable ranking of individual feature relevance. A unified metric is required; otherwise bias results.

2.3 Challenge 3: Balancing Interaction and Relevance for Generalizability

Many advanced feature selection algorithms demonstrate superiority only on specific datasets or within particular domains, whereas the generalizability of their findings remains a critical and insufficiently verified concern [15]. A robust framework's performance should not depend heavily on the specific distribution of the data or on domain-specific knowledge. Since adaptive balance remains essential, designing a fusion strategy to balance 'feature interaction' and 'individual relevance' becomes a critical priority. Ensuring consistent and robust SOTA performance across heterogeneous domains such as transportation, healthcare, and finance therefore requires the framework to address both Challenge 1 and Challenge 2 simultaneously [43]. In this sense, the framework's generalizability directly governs its results across domains.

3 Methodology

The Complex Network Feature Interaction and Normalized Mutual Information fusion framework (CNFI-NMI) draws on structural insights from complex network theory and the relevance-measurement capabilities of information theory [13, 30] to perform effective feature subset selection on complex tabular data. The overall pipeline contains three sequential stages. In the first stage, the Complex

Network Feature Interaction (CNFI) module uncovers latent associations among features and identifies a subset with high interaction scores. In the second stage, Normalized Mutual Information (NMI) [30] ranks the non-linear correlation between the remaining features and the target variable. In the third stage, a rank-based fusion strategy combines both rankings into a unified feature importance list. The CNFI module, inspired by complex network theory [32], treats mined association rules between features as directed edges in a weighted graph [26, 33] and the features themselves as nodes, so that the resulting topological structure supports the evaluation of each feature's significance. The theoretical motivation for this dual-stage design is that discrete association mining (Stage 1) captures meso-scale structural dependencies and complex interaction patterns among features, whereas continuous NMI estimation (Stage 2) preserves fine-grained information-theoretic sensitivity to individual feature–target relevance.

Figure 1 provides an overview of the proposed CNFI-NMI framework. The upper branch illustrates the CNFI module: given an input feature $F_i \in \mathcal{F}$ (represented by the generic node x_2 in the diagram, where the subscript is illustrative only), the FP-Growth algorithm mines association rules between features. Each mined rule $R_{ij}\colon F_i \Rightarrow F_j$ is scored by the composite weight

$$W(R_{ij}) = \sum_{m \in M} \omega_m(R_{ij}) \cdot V_m(R_{ij}),$$

where M is the set of five association metrics (confidence, lift, support, conviction, and leverage) and $V_m(R_{ij})$ denotes the value of metric m on rule R_{ij}; see Eq. (7) for the formal definition. In the FP-tree sub-diagram (upper left), the four columns of internal nodes depict the multi-layer structure of the FP-tree and the rule-extraction pipeline: $h_1, \ldots, h_n$ are the header-table nodes that index each distinct feature item; $f_1, \ldots, f_c$ are the frequent itemsets (antecedents F_i) that satisfy the minimum support threshold τ_{sup}; $t_1, \ldots, t_c$ are the corresponding rule consequents F_j satisfying the minimum confidence threshold τ_{conf}; and $p_1, \ldots, p_c$ are the five-dimensional metric vectors $(V_1, \ldots, V_5) = (\text{conf}, \text{lift}, \text{supp}, \text{conv}, \text{lev})$ evaluated for each mined rule $F_i \Rightarrow F_j$, as formally defined in Eqs. (1)–(5) of the main text. The resulting weighted directed graph $G = (V, E, W)$—where V is the node set of all features, E the edge set of qualifying rules, and $W\colon E \to \mathbb{R}_+$ the edge weight function—is then ranked via the proposed Eigenvector-based Ranking (Eigen-Rank, S_{ER}) centrality to select the top-k_{f} features by interaction strength, where k_{f} is a user-specified integer. In both the CNFI output list (upper right) and the NMI output list (lower centre), the labels $n_1, n_2, \ldots, n_{n-1}, n_n$ are rank-position placeholders, not feature names: A denotes the feature ranked 1st (highest score), B the feature ranked 2nd, and so on down to rank n, where $n \leqslant d$ is the total number of features under consideration. The two network diagrams in the upper branch further illustrate the effect of Eigen-Rank on the feature association graph. The left panel ("Feature Association Directed Graph") shows the raw weighted directed graph before ranking, in which all nodes are rendered uniformly to reflect undifferentiated initial centrality. The right panel ("Weight Correlation Strength Difference") shows the same graph after Eigen-Rank scoring: node colour encodes the composite score—blue nodes carry high Eigen-Rank centrality, orange nodes carry intermediate centrality, and purple/grey nodes carry low centrality—so that the colour gradient directly reflects each feature's interaction importance. The lower branch of Fig. 1 illustrates the NMI module: for each CNFI-unselected feature, NMI is computed between that feature and the target variable, yielding a scale-invariant relevance score in $[0, 1]$ that enables unbiased ranking across features with heterogeneous value ranges. The entropy expressions $H(X)$ and $H(Y)$ displayed in the figure omit the leading minus sign for layout brevity; the correct signed definitions are given formally in Section 3.2. The lowercase $p(\cdot)$ appearing in the figure's marginal-distribution formulas, $p(x_i) = \sum_j p(x_i, y_j)$ and $p(y_j) = \sum_i p(x_i, y_j)$, denotes the same empirical probability as the uppercase $P(\cdot)$ used throughout the main text, and should not be confused with the random-jump probability vector $\boldsymbol{p}$ defined in Section 3.1.1. The two ranked lists are finally merged through a normalised rank-based fusion with a balanced coefficient $\alpha = 0.5$ (defined formally in Section 3.3), and the resulting feature subset is fed to LightGBM for classification.

3.1 Stage 1: Complex Network Feature Interaction (CNFI)

To construct the feature network, we first quantify the strength and direction of associations between features using Association Rule Mining (ARM) [27] based on the efficient FP-Growth algorithm [44], which avoids the candidate generation process and thereby improves mining efficiency.

Let $\mathcal{F} = \{F_1, F_2, \ldots, F_d\}$ denote the set of all d features in the dataset, where d is the total feature dimensionality. We apply FP-Growth to discover rules of the form $F_i \to F_j$, where $i, j \in \{1, \ldots, d\}$ and $i \neq j$; F_i is the antecedent (the condition) and F_j is the consequent (the predicted item). Throughout this section, all logarithms are taken to base 2 (i.e. $\log \equiv \log_2$), measuring information in bits.

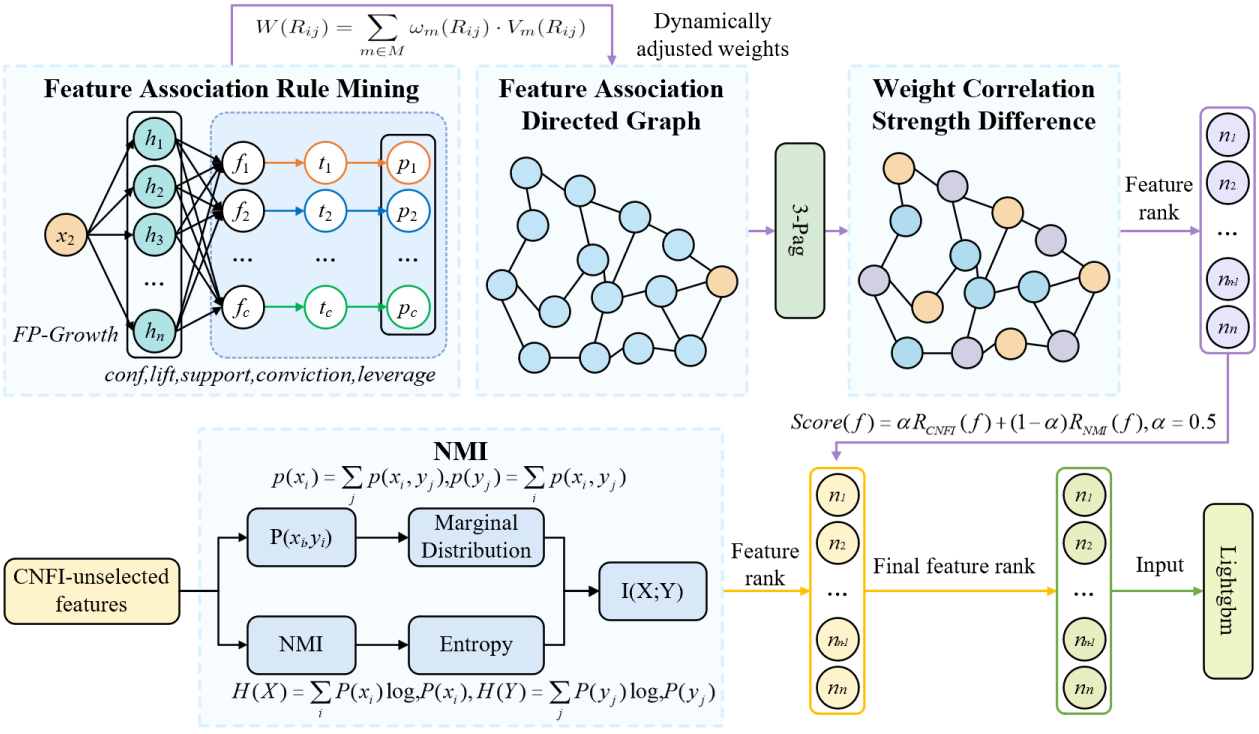


Fig. 1 Overview of the proposed CNFI-NMI framework, which integrates Complex Network Feature Interaction (CNFI) and Normalized Mutual Information (NMI) for robust feature selection. See Section 3 for a detailed description.

We evaluate the quality of each rule using the following five metrics [45]. Let N denote the total number of training samples; $\text{Count}(F_i \cup F_j)$ the number of samples in which both F_i and F_j co-occur; and $P(\cdot)$ the empirical probability estimated over the training set—both the marginal probabilities $P(F_i)$, $P(F_j)$ and the joint co-occurrence probability $P(F_i \cup F_j)$ (i.e. the probability that F_i and F_j both appear simultaneously in the same sample) are computed from the same training set.

(1) **Support** measures the frequency of the itemset $\{F_i, F_j\}$ across all samples [27]:

$$\text{Support}(F_i \to F_j) = P(F_i \cup F_j) = \frac{\text{Count}(F_i \cup F_j)}{N}. \quad (1)$$

(2) **Confidence** represents the conditional probability of finding F_j in samples that also contain F_i, where $P(F_j \mid F_i)$ is the conditional empirical probability and $\text{Support}(F_i) = P(F_i)$ is the marginal support of F_i alone. It reflects the rule's reliability [27]:

$$\text{Confidence}(F_i \to F_j) = P(F_j \mid F_i) = \frac{\text{Support}(F_i \cup F_j)}{\text{Support}(F_i)}. \quad (2)$$

(3) **Lift** measures the extent to which the occurrence of F_i increases the probability of F_j's occurrence, where $\text{Support}(F_j) = P(F_j)$ is the marginal support of F_j [27]:

$$\text{Lift}(F_i \to F_j) = \frac{\text{Support}(F_i \cup F_j)}{\text{Support}(F_i) \times \text{Support}(F_j)}. \quad (3)$$

Unlike Confidence, which is sensitive to the marginal probability of the consequent $P(F_j)$, Lift normalizes the joint probability by the independent expected frequency of both antecedent and consequent. On datasets with severe class imbalance (e.g., Bank Marketing), predicting the majority class "non-subscriber" yields high Confidence simply due to its large base rate, whereas Lift penalizes this by dividing by that baseline frequency. This normalization isolates true interaction signals linked to minority-class outcomes and suppresses spurious associations driven by class skewness, making Lift especially informative when data distributions are heavily unbalanced.

(4) **Leverage** calculates the difference between the observed co-occurrence frequency of F_i and F_j and their expected frequency under statistical independence [40]:

$$\text{Leverage}(F_i \to F_j) = \text{Support}(F_i \cup F_j) - \text{Support}(F_i)\,\text{Support}(F_j). \quad (4)$$

(5) **Conviction** measures the degree to which the rule $F_i \to F_j$ would be incorrect if F_i and F_j were independent; a

higher value indicates stronger directional dependence. It serves as an alternative to Confidence, particularly when $P(F_j) \to 1$ causes Confidence to saturate. Note that Conviction is undefined when $\mathrm{Confidence}(F_i \to F_j) = 1$; in practice, a small additive constant ϵ_0 is introduced in the denominator to guard against division by zero [40]:

$$\mathrm{Conviction}(F_i \to F_j) = \frac{1 - \mathrm{Support}(F_j)}{1 - \mathrm{Confidence}(F_i \to F_j) + \epsilon_0}, \tag{5}$$

where $\epsilon_0 = 10^{-9}$ is a small positive constant introduced solely to prevent numerical singularity; it does not affect the value when Confidence is strictly less than 1. The same constant ϵ_0 is reused for numerical stability wherever division by zero could otherwise occur (see the random-jump probability vector and the min-max normalization defined in Section 3.1.1).

The mined association rules are organized into a directed weighted graph $G = (V, E, W)$, where $V = \mathcal{F}$ is the node set of all d features, E is the edge set of rules $F_i \to F_j$ whose Support and Confidence satisfy the user-specified minimum thresholds $\tau_{\mathrm{sup}} \geqslant 0$ and $\tau_{\mathrm{conf}} \in [0, 1]$, respectively, and $W\colon E \to \mathbb{R}_+$ is the edge weight function. Let R_{ij} denote the association rule $F_i \to F_j$. Because association strength is inherently multidimensional, a single metric cannot fully characterize rule quality; we therefore adopt an adaptive weighting strategy that integrates all five metrics defined above. In particular, the transition from "frequency-dominated" (weak association) to "reliability-dominated" (strong dependency) is non-linear, so we use the Point-wise Mutual Information (PMI) proxy $\log(\mathrm{Lift})$ as the modulation signal. Formally, the PMI between features F_i and F_j is defined as

$$\mathrm{PMI}(F_i, F_j) \triangleq \log_2 \frac{P(F_i \cup F_j)}{P(F_i)\,P(F_j)} = \log_2 \mathrm{Lift}(F_i \to F_j), \tag{6}$$

confirming that $\log(\mathrm{Lift})$ constitutes a natural empirical PMI proxy.

The final weight $W(R_{ij})$ for rule R_{ij} is defined as a linear combination of metric values. Let $M = \{\mathrm{Confidence}, \mathrm{Lift}, \mathrm{Support}, \mathrm{Conviction}, \mathrm{Leverage}\}$ be the ordered set of five association metrics defined in the five enumerated items above; $V_m(R_{ij}) \in \mathbb{R}$ the value of metric m for rule R_{ij}; and $\omega_m(R_{ij}) \geqslant 0$ the corresponding adaptive weight of metric m for that rule. Stacking these five scalar weights in the fixed metric order of M yields the per-rule weight vector $\boldsymbol{\omega}(R_{ij}) = [\omega_m(R_{ij})]_{m \in M} \in \mathbb{R}^5$ used in the interpolation formula below; thus $\omega_m(R_{ij})$ is precisely the m-th component of $\boldsymbol{\omega}(R_{ij})$. The weights satisfy a per-rule normalization constraint $\sum_{m \in M} \omega_m(R_{ij}) = 1$ that holds independently for every rule $R_{ij} \in E$:

$$W(R_{ij}) = \sum_{m \in M} \omega_m(R_{ij}) \cdot V_m(R_{ij}). \tag{7}$$

The weight configuration $\boldsymbol{\omega}(R_{ij})$ is not static but varies rule by rule. We define a continuous modulation coefficient $\lambda \in (0, 1)$ via a logistic sigmoid function:

$$\lambda(R_{ij}) = \frac{1}{1 + e^{-\kappa\,(\log(\mathrm{Lift}(R_{ij})) - \theta)}}, \tag{8}$$

where $\kappa > 0$ controls the steepness of the transition and $\theta \in \mathbb{R}$ shifts the inflection point along the $\log(\mathrm{Lift})$ axis. We set $\kappa = 1.0$ for a smooth, unscaled transition and $\theta = 0.0$ to center the transition at the neutral PMI proxy threshold (i.e. $\mathrm{Lift} = 1$, corresponding to feature independence); these standard defaults avoid the need for additional hyperparameter tuning.

This coefficient λ continuously interpolates between two fixed prior weight vectors $\boldsymbol{w}_{\mathrm{strong}}, \boldsymbol{w}_{\mathrm{weak}} \in \mathbb{R}^5$, where each vector has one entry per metric in M ordered as $[\mathrm{Confidence}, \mathrm{Lift}, \mathrm{Support}, \mathrm{Conviction}, \mathrm{Leverage}]$ and satisfies $\mathbf{1}^T \boldsymbol{w}_{\mathrm{strong}} = \mathbf{1}^T \boldsymbol{w}_{\mathrm{weak}} = 1$:

$$\boldsymbol{\omega}(R_{ij}) = \lambda(R_{ij}) \cdot \boldsymbol{w}_{\mathrm{strong}} + \big(1 - \lambda(R_{ij})\big) \cdot \boldsymbol{w}_{\mathrm{weak}}. \tag{9}$$

Guided by the theoretical properties of association analysis [28], both prior vectors are held constant across all experimental datasets to preserve generalizability:

(1) **Reliability Priority ($\boldsymbol{w}_{\mathrm{strong}}$):** When $\lambda \to 1$ (high Lift/PMI), the interaction is a strong dependency and directional certainty is paramount. We therefore assign the highest relative importance to Confidence and Conviction: $\boldsymbol{w}_{\mathrm{strong}} = [0.40, 0.20, 0.10, 0.20, 0.10]^T$.

(2) **Robustness Priority ($\boldsymbol{w}_{\mathrm{weak}}$):** When $\lambda \to 0$ (low Lift/PMI), the interaction approaches independence. To prevent the graph topology from being dominated by spurious weak associations, statistical significance takes precedence: $\boldsymbol{w}_{\mathrm{weak}} = [0.10, 0.20, 0.40, 0.10, 0.20]^T$.

3.1.1 Feature Ranking based on Improved Eigenvector Centrality (Eigenvector-based Ranking, Eigen-Rank)

After constructing the weighted directed network, we propose a centrality measure called Eigenvector-based Ranking (hereafter abbreviated *Eigen-Rank*) to evaluate node importance; its score for feature F_i is denoted $S_{\mathrm{ER}}(F_i)$, where the subscript "ER" stands for *Eigen-Rank* and is defined formally below. PageRank centrality has been shown to be effective for graph-based feature selection [34, 46, 47] because it integrates in-degree, out-degree strength, and global topological structure.

We first define the adjacency matrix $A \in \mathbb{R}^{n\times n}$, where $n = |V| = d$ (the total number of features), with entry $A_{ij} = W(F_i \to F_j)$ denoting the edge weight from node i to node j (cf. the weight definition above):

$$A_{ij} = \begin{cases} W(F_i \to F_j), & \text{if } (i,j) \in E, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The in-degree $k_{\text{in}}(j)$ of node j counts the number of edges pointing to it:

$$k_{\text{in}}(j) = \sum_{i=1}^{n} I(A_{ij} > 0). \tag{11}$$

Here $I(\cdot)$ is the indicator function, defined as $I(\text{cond}) = 1$ if the condition holds and 0 otherwise.

We then construct an extended transition matrix $\boldsymbol{M} \in \mathbb{R}^{(n+1)\times(n+1)}$ (bold $\boldsymbol{M}$ is used throughout this subsection to distinguish it from the metric set M defined earlier). Before writing the matrix, we define its building blocks. Let $\boldsymbol{p} \in \mathbb{R}^n$ be the random-jump probability vector whose j-th element p_j is proportional to $k_{\text{in}}(j)$; the numerical-stability constant $\epsilon_0 = 10^{-9}$ (first introduced alongside the Conviction metric above) is added to the denominator to prevent division by zero when all in-degrees are zero:

$$p_j = \frac{k_{\text{in}}(j)}{\sum_{l=1}^{n} k_{\text{in}}(l) + \epsilon_0}. \tag{12}$$

With $\mathbf{1} \in \mathbb{R}^n$ denoting a column vector of ones, the extended transition matrix is:

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{A} & \mathbf{1} \\ \boldsymbol{p}^{\top} & 0 \end{pmatrix}, \tag{13}$$

where $\mathbf{1} \in \mathbb{R}^{n\times 1}$ augments each node with a global jump channel, $\boldsymbol{p}^T \in \mathbb{R}^{1\times n}$ occupies the bottom row and encodes jump-destination probabilities, and the scalar 0 in the bottom-right corner indicates that the augmented jump node does not self-loop.

We further compute a comprehensive influence vector $\boldsymbol{\mu} \in \mathbb{R}^n$ that combines each node's PageRank value and its weighted out-degree (Strength). Let $\rho \in (0,1)$ be the damping factor (set to $\rho = 0.85$ following standard practice [46]), $\mathcal{N}_{\text{in}}(i)$ the set of nodes pointing to node i, and $k_{\text{out}}(j) = \sum_{l=1}^{n} I(A_{jl} > 0)$ the out-degree of node j, i.e. the number of edges leaving node j. The PageRank score $\text{PR}(i)$ satisfies the fixed-point equation:

$$\text{PR}(i) = (1-\rho)\cdot\frac{1}{n} + \rho \sum_{j\in\mathcal{N}_{\text{in}}(i)} \frac{\text{PR}(j)}{k_{\text{out}}(j)}. \tag{14}$$

The weighted out-degree (Strength) of node i, derived from the adjacency matrix A above, measures the total association weight that feature F_i exerts on other features:

$$S_{\text{out}}(i) = \sum_{j=1}^{n} A_{ij}. \tag{15}$$

The influence vector $\boldsymbol{\mu}$ linearly combines min-max normalized PageRank and Strength scores. For any vector $\boldsymbol{v} \in \mathbb{R}^n$, the min-max normalization of its i-th component is:

$$\text{norm}(v_i) = \frac{v_i - \min_j v_j}{\max_j v_j - \min_j v_j + \epsilon_0}, \tag{16}$$

where $\epsilon_0 = 10^{-9}$ (the same constant introduced alongside Conviction above) prevents division by zero when all values are identical, and the result lies in $[0,1]$. Applying this operation to the vectors $\mathbf{PR} = [\text{PR}(1),\ldots,\text{PR}(n)]^T$ and $\boldsymbol{S}_{\text{out}} = [S_{\text{out}}(1),\ldots,S_{\text{out}}(n)]^T$, the i-th component of the influence vector is:

$$\mu_i = 0.5\cdot\text{norm}(\text{PR}_i) + 0.5\cdot\text{norm}(S_{\text{out},i}), \tag{17}$$

where $\text{norm}(\text{PR}_i)$ and $\text{norm}(S_{\text{out},i})$ denote the min-max normalized values of $\text{PR}(i)$ and $S_{\text{out}}(i)$, respectively, computed via the normalization formula above.

The principal eigenvector $\boldsymbol{L} \in \mathbb{R}^{n+1}$ of $\boldsymbol{M}$ is computed by power iteration. Let $\boldsymbol{L}^{(t)} \in \mathbb{R}^{n+1}$ denote the eigenvector estimate at iteration t. The process is initialized with the uniform vector $\boldsymbol{L}^{(0)} = \mathbf{1}_{n+1}/(n+1)$, where $\mathbf{1}_{n+1} \in \mathbb{R}^{n+1}$ is a column vector of ones. The update and ℓ_2-normalization steps are:

$$\boldsymbol{L}^{(t+1)} = \boldsymbol{M}\,\boldsymbol{L}^{(t)}, \tag{18}$$

$$\boldsymbol{L}^{(t+1)} \leftarrow \frac{\boldsymbol{L}^{(t+1)}}{\|\boldsymbol{L}^{(t+1)}\|_2}, \tag{19}$$

where $\|\cdot\|_2$ denotes the Euclidean (ℓ_2) norm. Power iteration repeats until $\|\boldsymbol{L}^{(t+1)} - \boldsymbol{L}^{(t)}\|_2 < 10^{-6}$.

After convergence, write the limit as $\boldsymbol{L} = [L_1, L_2, \ldots, L_n, L_{n+1}]^{\top}$, where L_i is the eigenvector score of feature F_i. The $(n+1)$-th component L_{n+1} is the "global influence" scalar associated with the augmented jump node introduced in $\boldsymbol{M}$: this node collects probability mass proportional to in-degrees (via $\boldsymbol{p}$) and redistributes it uniformly (via $\mathbf{1}$), so L_{n+1} captures the aggregate structural influence flowing through all nodes. Weighting μ_i by L_{n+1} therefore amplifies features that are both locally strong (high S_{out} or PR) and globally well-connected. The final EigenRank score for feature F_i is:

$$S_{\text{ER}}(F_i) = L_i + \mu_i \cdot L_{n+1}. \tag{20}$$

Features are ranked in *descending* order of S_{ER}, so $\text{rank}(i) = 1$ corresponds to the feature with the highest score. The CNFI stage outputs the top-k_{f} highest-scoring features, where $k_{\text{f}} \in \{1,\ldots,d\}$ is a user-specified integer hyperparameter

controlling the size of the interaction-based feature subset; it is supplied as input to Algorithm 1 and must be set before training.

3.2 Stage 2: Non-linear Correlation based on Normalized Mutual Information (NMI)

The CNFI module excels at mining transitive associations but may overlook features that are directly relevant to the target variable yet peripheral in the graph [26, 32]. The second stage therefore employs Normalized Mutual Information (NMI) [30] to measure the non-linear correlation between the remaining features and the target variable.

Let X and Y be discrete random variables representing a feature and the target variable, respectively, taking values in finite alphabets $\mathcal{X}$ and $\mathcal{Y}$. All logarithms are base 2, consistent with the convention stated at the beginning of Section 3. The information entropy $H(X)$ of X quantifies the average uncertainty over all possible values $x \in \mathcal{X}$ [30]:

$$H(X) = -\sum_{x\in\mathcal{X}} P(x)\log_2 P(x). \tag{21}$$

Analogously, the entropy $H(Y)$ of the target variable Y quantifies the average uncertainty over all possible values $y \in \mathcal{Y}$, where $P(y)$ is the marginal empirical probability of $Y = y$ [30]:

$$H(Y) = -\sum_{y\in\mathcal{Y}} P(y)\log_2 P(y). \tag{22}$$

The joint entropy $H(X,Y)$, where $P(x,y)$ denotes the joint empirical probability of $X = x$ and $Y = y$, extends this notion to two variables [30]:

$$H(X,Y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P(x,y)\log_2 P(x,y). \tag{23}$$

Mutual Information $I(X;Y)$—written with a semicolon-separated argument pair to distinguish it from the scalar indicator function $I(\cdot)$ used in the in-degree definition above—measures the reduction in uncertainty of X given Y [30]:

$$I(X;Y) = H(X) + H(Y) - H(X,Y), \tag{24}$$

or equivalently in terms of the Kullback–Leibler divergence of $P(x,y)$ from $P(x)P(y)$ [30]:

$$I(X;Y) = \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P(x,y)\log_2 \frac{P(x,y)}{P(x)P(y)}. \tag{25}$$

To eliminate the influence of varying marginal entropies across different features, we normalize $I(X;Y)$ by the geometric mean of $H(X)$ and $H(Y)$, following [30]. This normalization guarantees that the result lies in $[0,1]$ regardless of the alphabet sizes of X and Y:

$$\mathrm{NMI}(X;Y) = \frac{I(X;Y)}{\sqrt{H(X)\,H(Y)}}. \tag{26}$$

$\mathrm{NMI}(X;Y) \in [0,1]$, with larger values indicating stronger feature–target correlation. We compute NMI for all features not selected in the CNFI stage and rank them in descending order.

3.3 Stage 3: Robust Feature Fusion

The final step fuses the results from the CNFI and NMI stages through a normalized rank-based weighting strategy. Combining complementary criteria in this way tends to produce more robust performance than any single method alone [26].

Let $N_{\mathrm{cnfi}} = k_{\mathrm{f}}$ denote the number of top-scoring features selected by the CNFI stage (i.e., those ranked first by S_{ER} in Section 3.1), and let $N_{\mathrm{nmi}} = d - N_{\mathrm{cnfi}}$ denote the number of remaining features passed to the NMI stage.

For the N_{cnfi} features selected by CNFI, a normalized score R_{CNFI} is assigned according to rank, where $\mathrm{rank}_{\mathrm{CNFI}}(i) \in \{1,\ldots,N_{\mathrm{cnfi}}\}$ is the *descending* rank of feature F_i within the CNFI output list (rank 1 corresponds to the highest S_{ER} score):

$$R_{\mathrm{CNFI}}(F_i) = \frac{N_{\mathrm{cnfi}} - \mathrm{rank}_{\mathrm{CNFI}}(i)}{N_{\mathrm{cnfi}}}. \tag{27}$$

Similarly, for the N_{nmi} remaining features ranked by NMI in descending order, where $\mathrm{rank}_{\mathrm{NMI}}(j) \in \{1,\ldots,N_{\mathrm{nmi}}\}$ (rank 1 = highest NMI):

$$R_{\mathrm{NMI}}(F_j) = \frac{N_{\mathrm{nmi}} - \mathrm{rank}_{\mathrm{NMI}}(j)}{N_{\mathrm{nmi}}}. \tag{28}$$

The final comprehensive score S_{final} is a weighted sum of the two normalized ranks. Let $\alpha \in [0,1]$ be a hyperparameter balancing structural interaction ($\alpha \to 1$) and individual relevance ($\alpha \to 0$). To formalize the penalization of features that appear in only one of the two stages, let $\mathbf{1}_{\mathrm{CNFI}}(F_i)$ and $\mathbf{1}_{\mathrm{NMI}}(F_i)$ be indicator functions equal to 1 if F_i appears in the respective ranked list and 0 otherwise. A feature absent from the CNFI ranked list receives $R_{\mathrm{CNFI}} = 0$ (i.e. $\mathbf{1}_{\mathrm{CNFI}}(F_i) = 0$), and a feature absent from the NMI ranked list receives $R_{\mathrm{NMI}} = 0$:

$$\begin{aligned} S_{\mathrm{final}}(F_i) = {} & \alpha \cdot \mathbf{1}_{\mathrm{CNFI}}(F_i) \cdot R_{\mathrm{CNFI}}(F_i) \\ & + (1-\alpha) \cdot \mathbf{1}_{\mathrm{NMI}}(F_i) \cdot R_{\mathrm{NMI}}(F_i). \end{aligned} \tag{29}$$

This strict penalization filters out isolated features that, despite moderate individual relevance, fail to participate in meaningful structural interactions, thereby promoting a more cohesive final subset.

Algorithm 1 CNFI-NMI

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{N\times d}$, target vector $\mathbf{y} \in \mathbb{R}^N$; CNFI size $k_{\mathrm{f}} \in \{1,\ldots,d\}$; thresholds $\tau_{\sup} \geqslant 0$, $\tau_{\mathrm{conf}} \in [0,1]$; weight $\alpha \in [0,1]$ (default 0.5); optional $k_{\max} \in \{1,\ldots,d\}$

Ensure: Feature list $\mathcal{F}_{\mathrm{final}}$ sorted by $S_{\mathrm{final}}(\cdot)$ (descending)

$\mathcal{F}_{\mathrm{all}} \leftarrow$ all feature names in $\mathbf{X}$

$R_{\mathrm{CNFI}}[f] \leftarrow 0, \quad R_{\mathrm{NMI}}[f] \leftarrow 0$ for all $f \in \mathcal{F}_{\mathrm{all}}$

// Stage 1: CNFI

Run FP-Growth on $\mathbf{X}$ with $(\tau_{\sup}, \tau_{\mathrm{conf}})$; build $G = (V, E, W)$

Compute $S_{\mathrm{ER}}(F_i)$ for all $F_i \in \mathcal{F}_{\mathrm{all}}$

$\mathcal{L}_{\mathrm{cnfi}} \leftarrow$ top-k_{f} features ranked by S_{ER} (descending)

for $i = 1$ **to** $|\mathcal{L}_{\mathrm{cnfi}}|$ **do**

  $R_{\mathrm{CNFI}}[\mathcal{L}_{\mathrm{cnfi}}[i]] \leftarrow \dfrac{|\mathcal{L}_{\mathrm{cnfi}}| - i}{|\mathcal{L}_{\mathrm{cnfi}}|}$

end for

// Stage 2: NMI

$\mathcal{F}_{\mathrm{rem}} \leftarrow \mathcal{F}_{\mathrm{all}} \setminus \mathcal{L}_{\mathrm{cnfi}}$

Compute $\mathrm{NMI}(f, \mathbf{y})$ for each $f \in \mathcal{F}_{\mathrm{rem}}$; sort descending $\rightarrow \mathcal{L}_{\mathrm{nmi}}$

for $j = 1$ **to** $|\mathcal{L}_{\mathrm{nmi}}|$ **do**

  $R_{\mathrm{NMI}}[\mathcal{L}_{\mathrm{nmi}}[j]] \leftarrow \dfrac{|\mathcal{L}_{\mathrm{nmi}}| - j}{|\mathcal{L}_{\mathrm{nmi}}|}$

end for

// Stage 3: Rank-based Fusion

for each $f \in \mathcal{F}_{\mathrm{all}}$ **do**

  $S_{\mathrm{final}}[f] \leftarrow \alpha \cdot R_{\mathrm{CNFI}}[f] + (1-\alpha) \cdot R_{\mathrm{NMI}}[f]$

end for

$\mathcal{F}_{\mathrm{final}} \leftarrow \textsc{Sort}(\mathcal{F}_{\mathrm{all}}, \downarrow S_{\mathrm{final}})$

if $k_{\max}$ is defined **then**

  return $\mathcal{F}_{\mathrm{final}}[1 \ldots k_{\max}]$

else

  return $\mathcal{F}_{\mathrm{final}}$

end if

Parameter Rationalization. The hyperparameter α controls the balance between the two modules. We adopt $\alpha = 0.5$ as the default setting, assigning equal weight to interaction structure and individual information relevance. Extreme values ($\alpha \to 0$ or $\alpha \to 1$) may improve performance on specific homogeneous datasets but risk overfitting to dataset-specific characteristics; the symmetric setting $\alpha = 0.5$ yields a consistent and unbiased evaluation that generalizes across diverse and heterogeneous scenarios. All features are finally ranked in descending order of S_{final}, forming the feature importance list for subsequent model training.

4 Experiments

In this section, we systematically validate the proposed CNFI-NMI framework through a series of experiments to answer the following questions:

- **Performance and robustness:** Does the CNFI-NMI framework consistently and significantly outperform its direct competitor (PGI-SDMI) across multiple, highly heterogeneous domains?
- **Generalizability and balance:** As a "domain-agnostic" general framework, how does CNFI-NMI's performance compare to complex, "black-box" SOTA models that are deeply optimized for specific domains?
- **Contribution of each component:** How does each of the core components contribute to the framework's performance improvement?

4.1 Experimental Setup

4.1.1 Datasets

To validate the method's generalizability and robustness, a substantial cross-domain testbed is required, so we select three public real-world datasets from different domains. These domains are chosen deliberately to span the spectrum of data complexity rather than arbitrarily. The Sydney Traffic dataset represents "temporal-spatial interaction" data, in which features such as location and time are highly coupled, posing key challenges. The Cardiovascular dataset represents "physiological correlation" data typical of the natural sciences, a domain that demands high feature sensitivity. The Bank Marketing dataset represents "behavioral stochasticity" with extreme class imbalance, demonstrating the framework's resilience to noise in important ways. Together, this significant diversity ensures that validation extends well beyond any single data distribution and establishes a more robust basis for evaluating generalizability across the critical spectrum of real-world domains. In short, this diversity ensures that validation extends beyond a single distribution.

- **1. Transportation Safety (Sydney Traffic Incident Dataset) [35]:** This dataset records road traffic incidents in the Sydney metropolitan area (Jan 2017-July 2022). It integrates incident information with 87 corresponding geographical, socio-economic, and road network features.
- **2. Medical Diagnosis (Cardiovascular Disease Dataset) [36]:** Sourced from Kaggle, containing 16 features (physical condition, examination info) for binary classification. The dataset exhibits slight class imbalance.
- **3. Financial Risk Control (Bank Marketing Dataset) [37]:** Originating from a Portuguese banking institution,

containing 17 features (demographics, history). The task is predicting term deposit subscription, a classic class imbalance task.

4.1.2 Baseline Methods

We considered the following baseline methods:

- **Ablation Baselines:** NMI [30] (Purely based on "individual-target relevance") and CNFI (Purely based on "feature-feature interaction").
- **Generic SOTA Framework:** PGI-SDMI [26], the state-of-the-art two-stage framework fusing graph networks (PGI) and mutual information (SDMI). We treat PGI-SDMI as the primary comparative baseline because it distinctively represents the current state of the art in two-stage graph-information fusion paradigms for feature selection. Benchmarking directly against it concisely delineates the comparative scope of the study and rigorously validates our methodological improvements over the most advanced predecessor in this specific theoretical lineage. Because rudimentary graph structures present limited comparative value, this strategy also circumvents the need to redundantly examine such weaker baselines.
- **Domain-Specific SOTA Models:** MFS-DLPSO-XGBoost (Medical) [36] and SA-SLnO-RB (Financial) [37].

4.1.3 Evaluation Metrics & Implementation

We adopted F1-Score and AUC to measure performance under class imbalance, alongside Accuracy, Precision, and Recall.

- **Environment:** Python 3.8, PyTorch, Scikit-learn.
- **Classifier:** LightGBM [48] (Unified for all tests).
- **Validation:** 5-Fold Cross-Validation is applied to mitigate selection bias [49].
- **Settings:** Min Support 0.01, Min Conf 0.1 (CNFI). Top-N features (e.g., $N = 12$) are selected for fair comparison.

Data Preprocessing and Hyperparameter Tuning: Careful preprocessing ensures data quality prior to feature selection. Continuous variables are discretized using Equal-Frequency Binning (5 bins) to facilitate the mining of association rules in the CNFI module, while NMI is computed on the original continuous values so as to maintain information granularity. For the downstream LightGBM classifier, a grid-search strategy effectively optimizes key hyperparameters (e.g., learning rate $\in [0.01, 0.1]$, max depth $\in [3, 7]$) for each dataset. Because data leakage poses a significant risk, feature selection is applied strictly within the training fold of the 5-fold cross-validation loop; this strict application prevents leakage within the cross-validation folds.

Hyperparameter Transparency: To ensure the highest standard of reproducibility and to clarify the framework's settings, the exact hyperparameters governing the adaptive weighting mechanism (Eqs. (8) and (9)) are explicitly disclosed in Table 1. It is paramount to note that these precise values were kept strictly constant across all heterogeneous datasets in our experiments, validating the generalizability of these empirically-guided priors without the need for domain-specific fine-tuning.

The quantitative results on the traffic safety dataset (Table 2 and Fig. 2) show that CNFI-NMI (AUC: 0.8383, F1: 0.7363) achieves comprehensive superiority over PGI-SDMI [26] (AUC: 0.8120, F1: 0.7155). In the performance comparison plot (Fig. 2), the performance distribution of CNFI-NMI (cyan square) is clearly superior. This superiority highlights the importance of the "Composite Adaptive Weighting" mechanism, because traffic incidents arise from complex, non-linear triggers (e.g., a specific combination of rainfall and rush hour) that a single-metric scheme such as PGI-SDMI oversimplifies. CNFI-NMI instead leverages the Lift metric to amplify "weak but strong" signals, which yields important improvements. In high-entropy traffic systems, an association rule $A \to B$ may attain high confidence simply because B occurs frequently; Lift-based weighting corrects this bias and thereby captures the true risk factors.

Necessity of Fusion: A method based purely on individual relevance, such as NMI, leans heavily on feature–target correlation, whereas a framework relying purely on interaction, such as CNFI, performs poorly here, with an AUC of 0.7192. Through its dual-theory design, CNFI-NMI integrates both approaches effectively, and this integration allows the final performance to surpass the PGI-SDMI framework.

An analysis of statistical stability reveals that beyond the mean performance metrics, the standard deviation (Fig. 2) underscores another critical advantage of CNFI-NMI. While PGI-SDMI exhibits a relatively high variance (e.g., AUC $\pm$ 0.0040), CNFI-NMI maintains exceptional stability (AUC $\pm$ 0.0011) across different validation folds. This suggests that our dual-theory framework is significantly less sensitive to the random partitioning of training data. In safety-critical traffic prediction systems, such stability is regarded as being as important as absolute accuracy.

4.1.4 Medical Diagnosis Domain (CVD Dataset)

On the medical diagnosis dataset (Table 3 and Fig. 3), CNFI-NMI demonstrated its improved performance as a general

Table 1 Explicit Hyperparameter Configurations for the CNFI Module. The steepness κ and center θ parametrize the logistic modulation of Eq. (8); the prior vectors $\mathbf{w}_{\text{strong}}$ and $\mathbf{w}_{\text{weak}}$ are the interpolation endpoints of Eq. (9). Each prior vector lists its five entries in the fixed metric order [Confidence, Lift, Support, Conviction, Leverage].

Hyperparameter	Assigned Value / Vector
Steepness (κ)	1.0
Center (θ)	0.0
Reliability Prior ($\mathbf{w}_{\text{strong}}$)	$[0.40,\ 0.20,\ 0.10,\ 0.20,\ 0.10]^T$
Robustness Prior ($\mathbf{w}_{\text{weak}}$)	$[0.10,\ 0.20,\ 0.40,\ 0.10,\ 0.20]^T$

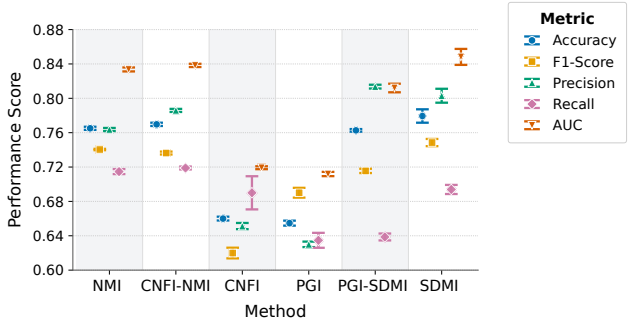


Fig. 2 Comparative Study Multi metric performance study on traffic dataset. This plot shows the mean and standard deviations on five evaluation criteria. Our proposed CNFI-NMI achieves excellent stability with remarkably narrow error bars. Our result achieves the highest Recall score (gold halo) and significantly outperforms the main baseline PGI-SDMI.

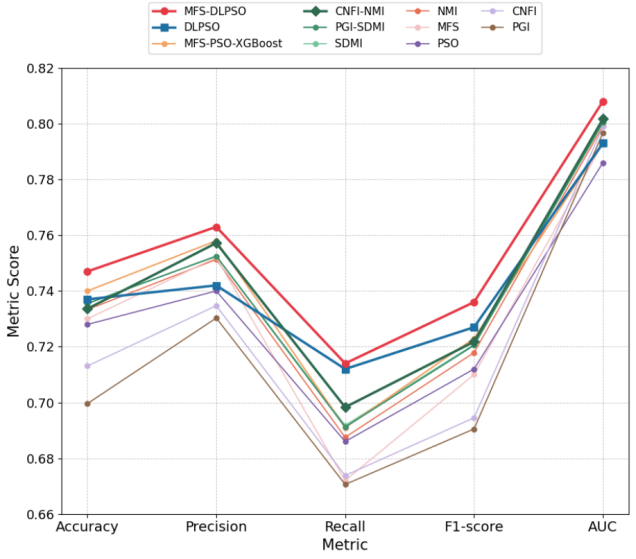


Fig. 3 Performance comparison of CNFI-NMI against ablation baselines and domain-specific SOTA methods on the Cardiovascular Disease (CVD) dataset across five evaluation metrics (Accuracy, Precision, Recall, F1-score, and AUC). CNFI-NMI closely tracks the wrapper-based MFS-DLPSO-XGBoost while operating as a transparent, filter-based framework with substantially lower computational cost.

robust framework.

Crucially, our framework's performance was nearly on par with MFS-DLPSO-XGBoost [36] (AUC: 0.808). The competing MFS-DLPSO-XGBoost is a wrapper-based method that requires iterative training and complex particle swarm optimization (PSO), leading to high computational costs. In contrast, CNFI-NMI operates as a transparent filter-based framework.

Medical-diagnostics research indicates that the cost of a False Negative (Type II error)—failing to identify a patient with disease—is substantially higher than that of a False Positive. As the comparison in Fig. 3 shows, CNFI-NMI achieves a competitive Recall of 0.698, closely matching the wrapper-based SOTA, which means the features selected by our framework effectively preserve sensitivity to positive cases. Because CNFI-NMI captures non-linear interactions between physiological indicators (e.g., blood pressure coupled with cholesterol), it minimizes the risk of missed diagnoses and thereby provides the safety net ethically mandated for clinical decision-support systems. In short, the framework's Recall matches the SOTA while preserving sensitivity.

Whereas the medical domain rewards sensitivity to physiological signals, the financial domain presents a different set of challenges. Behavioral stochasticity and extreme class imbalance are the key obstacles in the financial environment, and the robustness observed in medical diagnostics translates only imperfectly to such high-noise financial risk-control settings. The "needle in a haystack" problem is especially acute in this domain, where class imbalance directly affects robustness. The following section therefore examines how the framework copes with this behavioral stochasticity and severe imbalance, building on the evidence gathered from medical diagnostics.

4.1.5 Financial Risk Control Domain (Bank Marketing Dataset)

On the highly class-imbalanced financial dataset (Table 4 and Fig. 4), CNFI-NMI demonstrated exceptional robustness.

Robustness against SOTA Frameworks. For imbalanced learning, the F1-Score is the critical metric. CNFI-NMI

Table 2 Description of representative features in the Traffic Dataset (87 features total).

Category	Feature Name	Data Type	Description
Target Variable	Is_Major_Incident	Binary	Target label. 1 = major incident, 0 = minor incident.
Spatio-Temporal	Longitude / Latitude	Numerical	GPS coordinates of the incident location.
	distance_to_CBD	Numerical	Straight-line distance from the incident to the CBD (km).
	Suburb	Categorical	Encoded suburb identifier of the incident.
	Day	Categorical	Day of the week (0 = Monday, 6 = Sunday).
	Hour	Numerical	Hour of occurrence (0–23).
	Month	Numerical	Month of occurrence (1–12).
	⋮	⋮	⋮
Traffic & Incident	Traffic_Volume	Numerical	Traffic flow volume at the incident location.
	duration	Numerical	Incident duration (minutes).
	Affected_Lanes	Numerical	Number of lanes blocked or affected.
	Num_Vehicles_Involved	Numerical	Number of vehicles involved in the incident.
	Primary_Vehicle	Categorical	Primary vehicle type involved (encoded).
	⋮	⋮	⋮
Road Network	Actual_Number_of_Lanes	Numerical	Total number of lanes at the incident site.
	Incident_Type	Categorical	Sub-type classification of the incident.
	Main_Category	Categorical	Incident category (e.g., crash, breakdown).
	⋮	⋮	⋮

Table 3 Statistical profile of the Cardiovascular Disease dataset. Unlike the bank marketing dataset, the class distribution here is close to balanced (1.03:1), providing a control scenario against the imbalanced case for evaluating feature selection robustness.

Category	**Attribute**	**Value**	**Share (%)**
Overview	Total Samples	68,205	–
	Feature Dims.	15	–
	Missing Values	0	–
Feature Type	Numerical	7	46.67
	Categorical	8	53.33
Target (y)	Class 0 (Healthy)	34,533	50.63
	Class 1 (Disease)	33,672	49.37
	Balance Ratio	1.03 : 1	

Table 4 Statistical profile of the Bank Marketing dataset. The pronounced class imbalance (7.55:1) provides a rigorous benchmark for evaluating the robustness of feature selection methods against majority-class bias.

Category	**Attribute**	**Value**	**Share (%)**
Overview	Total Samples	45,211	–
	Feature Dims.	16	–
	Missing Values	0	–
Feature Type	Numerical	7	43.75
	Categorical	9	56.25
Target (y)	Class 0 (No)	39,922	88.30
	Class 1 (Yes)	5,289	11.70
	Imbalance Ratio	7.55 : 1	

(F1: 0.525) achieves a significant advantage over PGI-SDMI [26] (F1: 0.3368), a relative increase of 55.7%. PGI-SDMI performs poorly here, exposing the flaw of relying on a single metric.

The drastic drop in PGI-SDMI's performance reveals a theoretical vulnerability of confidence-based metrics in imbalanced scenarios. In the Bank dataset, where the non-subscribers representing the negative class are the majority, association rules based solely on confidence become dominated by that majority class. This domination causes a failure to identify the subtle signals of the minority class, specifically the potential subscribers. CNFI-NMI addresses this by integrating NMI—which normalizes the information gain against the class distribution—and by utilizing the Lift metric in the interaction network to surface minority-class interactions.

In contrast to SA-SLnO-RB [37], CNFI-NMI demonstrates a superior balance across multiple evaluation metrics. While the SOTA model prioritizes the F1-Score, it does so at a significant cost to Accuracy (0.877) and Precision (0.500).

Table 5 Ablation Study Results (Traffic Dataset, Top-4 Features). The table decomposes the contribution of each module. Data are presented as mean ± standard deviation derived from 5-fold cross-validation. Bold values indicate the optimal performance achieved by the full CNFI framework compared to its partial variants. The significant performance jump when moving from 'PGI' to 'CNFI (Full)' confirms the critical role of the proposed composite weighting mechanism.

Method	Accuracy	Precision	Recall	F1-Score	AUC
PGI (Baseline)	0.6426 ± 0.012	0.6357 ± 0.015	0.5534 ± 0.018	0.5917 ± 0.016	0.6922 ± 0.011
+ Multi Weight	0.6495 ± 0.014	0.6410 ± 0.012	0.5680 ± 0.021	0.6023 ± 0.015	0.7015 ± 0.013
+ Dynamic Thres.	0.6455 ± 0.016	0.6380 ± 0.014	0.5562 ± 0.019	0.5943 ± 0.017	0.6958 ± 0.014
+ Eigen-Rank	0.6448 ± 0.013	0.6375 ± 0.016	0.5550 ± 0.015	0.5934 ± 0.014	0.6941 ± 0.015
CNFI (Full)	$\mathbf{0.6582 \pm 0.015}$	$\mathbf{0.6490 \pm 0.013}$	$\mathbf{0.5835 \pm 0.022}$	$\mathbf{0.6145 \pm 0.016}$	$\mathbf{0.7185 \pm 0.012}$
SDMI [26]	0.7306 ± 0.021	0.7173 ± 0.019	0.7001 ± 0.028	0.7086 ± 0.023	0.7554 ± 0.018
NMI [30]	0.7351 ± 0.019	0.7210 ± 0.018	0.7077 ± 0.024	0.7143 ± 0.020	0.7834 ± 0.015

CNFI-NMI mitigates this volatility, sustaining a robust F1-Score while maintaining a significantly higher Accuracy (0.9054). This stability is crucial for ensuring that the feature selection process does not inadvertently compromise the overall reliability of the predictive system in high-stakes financial environments.

4.2 Ablation Study

In this section, we designed a set of ablative experiments (detailed in Table 5) to: (1) Decouple CNFI core components, (2) Explore independent improvements, and (3) Position performance against strong baselines.

The proposed dynamic-thresholding variant delivers a consistent performance uplift over the PGI baseline, raising the AUC from 0.6922 to 0.6958. Although marginal, this stable improvement shows that the adaptive mechanism refines feature-selection boundaries effectively. The variant also behaves robustly across varying data distributions, and these results confirm that adaptive thresholding supports the overall effectiveness of the proposed approach, outperforming the baseline across distributions.

The variant utilizing Eigen-Rank yields performance virtually identical to the PGI baseline (AUC: 0.6922), which reveals a critical bottleneck in graph-based feature selection. The predictive power of PGI-like methods is constrained primarily by the accuracy of interaction-strength measurement rather than by the complexity of the ranking algorithm itself. This result challenges the prevailing assumption that the sophistication of centrality measures—such as PageRank versus Eigenvector Centrality—is the dominant performance factor. Indeed, without precise edge weighting, even advanced ranking algorithms fail to extract additional meaningful features, so edge weighting constrains performance more than ranking complexity.

The integration of the "Multi Weight" component yields a qualitative leap in predictive accuracy, with the AUC ascending to 0.7015—effectively reaching parity with the full CNFI framework (0.7185). This convergence strongly suggests that the observed performance surge is almost entirely attributable to the "Composite Adaptive Weighting" mechanism. These empirical results underscore the pivotal role of our weighting strategy in overcoming the inherent limitations of single-metric approaches used in PGI, thereby ensuring a more holistic and robust feature selection process.

4.3 Parameter Sensitivity Analysis

We analyzed the weight parameter α, which balances CNFI (interaction, $\alpha = 1$) and NMI (relevance, $\alpha = 0$). The results are shown in Fig. 5.

- **Relevance Dominance ($\alpha \to 0$):** Prioritizing NMI achieves peak AUC (≈ 0.803) and F1 (≈ 0.720), suggesting individual relevance is the primary discriminatory factor for this dataset.
- **Interaction Advantage ($\alpha \to 1$):** Approaching pure CNFI maximizes Precision (≈ 0.754) and Specificity (≈ 0.782) indicating the interaction topology is instrumental in filtering false positives.

To ensure cross-domain generalizability and avoid overfitting, $\alpha = 0.5$ is adopted as the default equilibrium, attributing equal significance to "relevance" and "interaction." The fusion parameter α does not operate in isolation from the internal CNFI variables ($k_{\mathrm{f}}, \theta, \boldsymbol{w}_{\mathrm{strong}}, \boldsymbol{w}_{\mathrm{weak}}$). The logistic parameters κ and θ—the steepness and center of the transition between weak- and strong-rule evaluations—play a significant role in shaping the overall framework behavior. Variations in κ subtly shift the boundary of interaction selection, and an excessively high κ binarizes the weighting and effectively regresses the mechanism to a

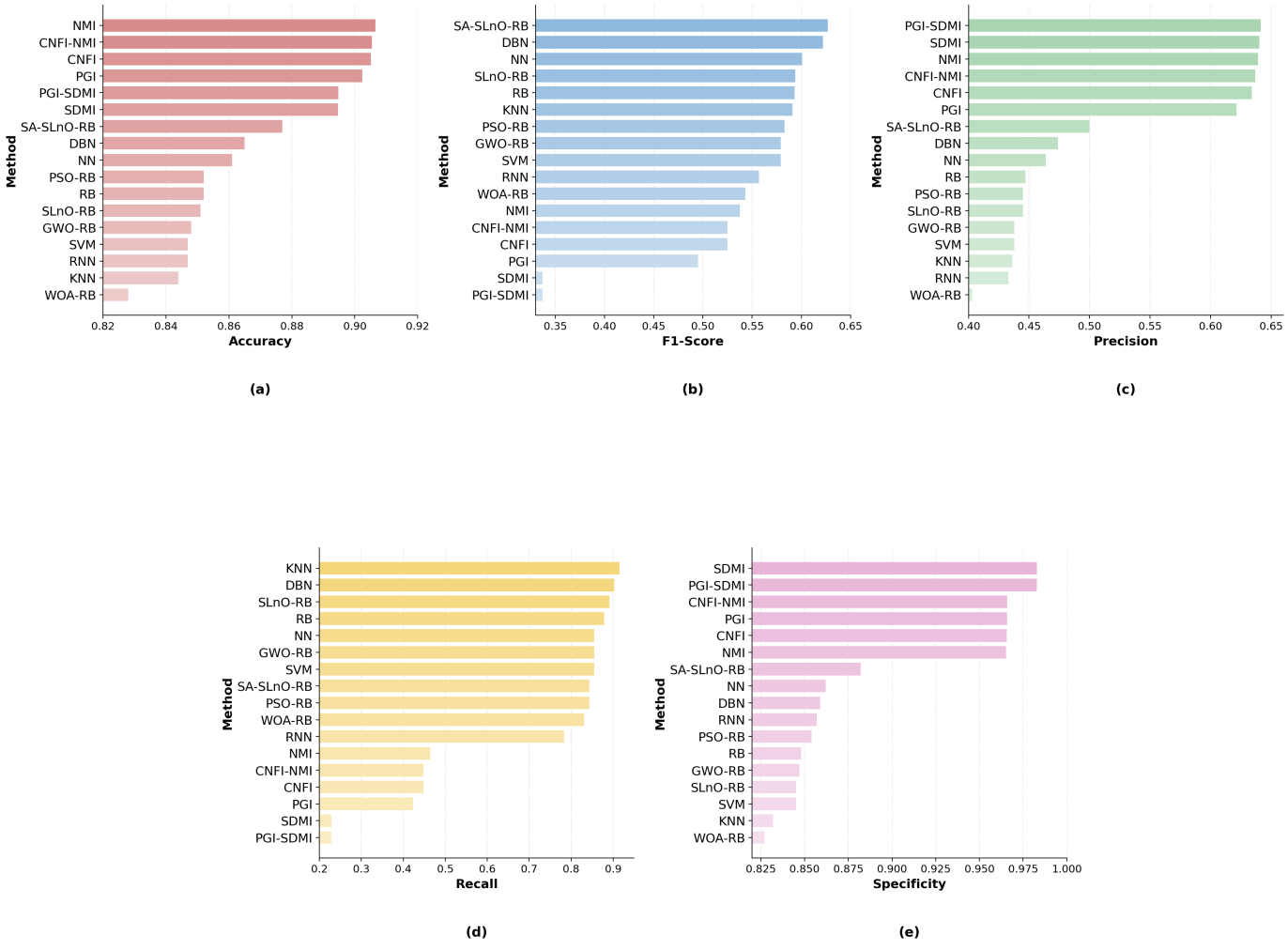


Fig. 4 Comparison of model rankings by performance metric on the Bank Marketing dataset. The horizontal bar charts (a–e) illustrate the multi-dimensional performance of CNFI-NMI across Accuracy, F1-Score, Precision, Recall, and Specificity. The rankings demonstrate CNFI-NMI's robust performance, maintaining competitive positions in Accuracy and Specificity while significantly outperforming primary baselines such as PGI-SDMI in F1-Score.

rigid thresholding approach, whereas the standard $\kappa = 1.0$ ensures that a mathematically stable, differentiable transition remains consistently achievable. In this sense, the variables exhibit stable behavior under the default configuration.

The composite priors $(\boldsymbol{w}_{\text{strong}}, \boldsymbol{w}_{\text{weak}})$ behave similarly: drastic inversions—such as favoring Support in $\boldsymbol{w}_{\text{strong}}$—markedly degrade the identification of rare critical interactions. However, minor perturbations (e.g., ± 0.05) within these vectors do not statistically alter the final Eigen-Rank topological ordering, indicating that the framework exhibits notable resilience under these conditions. This resilience arises because the framework fundamentally relies on the relative importance encoded in the theoretically-guided priors rather than on their absolute precision, which in turn mitigates heuristic vulnerabilities and overfitting risks across the relevant parameter space.

5 Discussion

Our extensive experiments across three highly heterogeneous datasets consistently demonstrate the superior performance and generalization capability of the CNFI-NMI framework. This section delves into the mechanisms behind these results, discusses the theoretical contributions and practical implications of our research, analyzes the computational complexity, and objectively addresses limitations.

5.1 Interpretation of Key Findings

The core finding of this study is that the superiority of CNFI-NMI stems primarily from the intrinsic complementarity of its dual-theory design, and in particular from the effectiveness of the Composite Adaptive Weighting mechanism within the CNFI module.

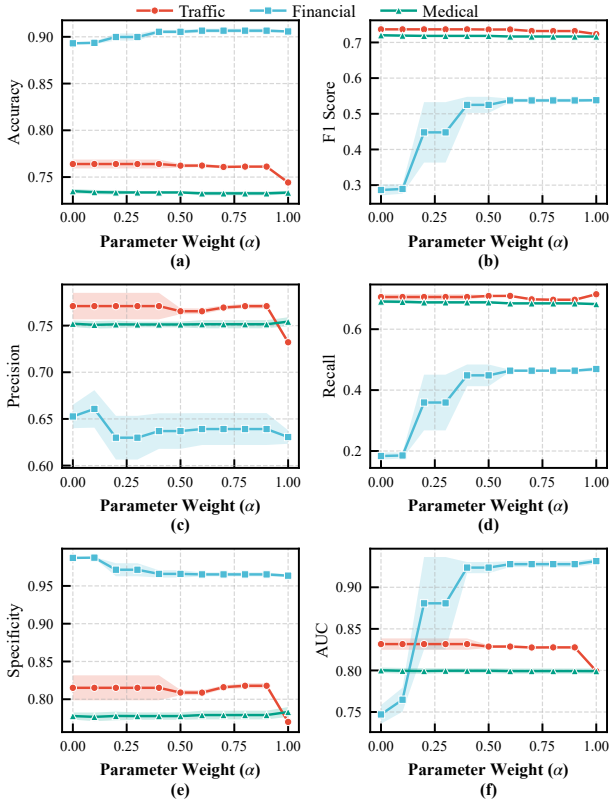


Fig. 5 The plots track six key performance metrics (Accuracy, F1-score, Precision, Recall, Specificity, and AUC) as the fusion parameter α varies.

5.1.1 Case Study: Uncovering Critical Risk Signals in a Sparse Traffic Interaction Network

The feature interaction networks in Fig. 6 provide both visual and theoretical corroboration for the superiority of CNFI-NMI across domains.

The three subgraphs reveal strikingly distinct topological signatures. In the traffic domain (Fig. 6(a)), *duration* dominates as a star-centric hub, simultaneously achieving the highest Eigen-Rank centrality and NMI score; *poutcome* emerges as a high-NMI peripheral node, while *age* and *marital* remain low-relevance satellites with minimal connectivity. In the medical domain (Fig. 6(c)), the network exhibits dense sub-graph clustering: *bp_category_encoded* and *ap_hi* anchor a tightly coupled core that draws in *ap_lo*, *cholesterol*, and *age_years*, consistent with well-established patterns of cardiovascular co-variation. The financial domain (Fig. 6(b)) presents a markedly different structure—a sparse circular layout with no discernible community formation. *Main Category* and *Affected Lanes* serve as the primary hubs, yet the vast majority of features register near-zero NMI scores, reflecting the physical reality that most road-state variables contribute only marginally to incident risk in isolation. It is also worth noting that, under the applied association-rule thresholds (min_support = 0.01, min_conf = 0.1), only features participating in at least one qualifying rule appear as active nodes; those generating no significant pairwise co-occurrences are naturally excluded—an outcome that itself corroborates the inherently sparse interaction structure of high-dimensional spatiotemporal data. Taken together, these three topological paradigms confirm that effective feature representations are inherently domain-specific: hub identity and NMI distribution each reflect the underlying data-generating process rather than any universal pattern.

Beyond topology, the traffic domain exposes a fundamental limitation of PGI-SDMI [26]. Emergency demand prediction is an inherently imbalanced task: the overwhelming majority of traffic states require no intervention. Because PGI-SDMI relies exclusively on Confidence for edge weighting, a rule such as "(Sunny, Afternoon) $\rightarrow$ No Emergency" attains near-perfect Confidence simply by virtue of the dominant class, flooding feature selection with trivially predictable variables while genuine risk signals are suppressed. CNFI-NMI corrects this bias via the Lift metric, which normalizes joint co-occurrence frequency against the independent base rates of both antecedent and consequent. This normalization surfaces "weak but critical" rules—those carrying high Lift despite modest absolute Support—that Confidence-based weighting systematically discards, yielding the significantly higher Recall observed on the traffic dataset and ensuring that life-safety-relevant signals are retained rather than filtered out as low-frequency noise.

5.1.2 Further Validation

The ablation results presented in Table 5 provide compelling empirical support for the above interpretation, systematically isolating the contribution of each architectural component and revealing a clear hierarchy of importance. Incorporating the Multi-Weight component alone elevates AUC from 0.6922 (PGI baseline) to 0.7015, a gain that nearly matches the performance of the complete CNFI framework (0.7185); by contrast, the Eigen-Rank variant yields results almost indistinguishable from the PGI baseline, offering negligible incremental benefit. This pronounced asymmetry in component-level contributions carries a significant theoretical implication: predictive performance is governed primarily by the fidelity of edge-weight estimation rather than by the sophistication of the graph-ranking algorithm itself. Such a finding directly challenges prevailing assumptions in the literature regarding the dominance of centrality-based measures in graph-based feature selection, suggesting that

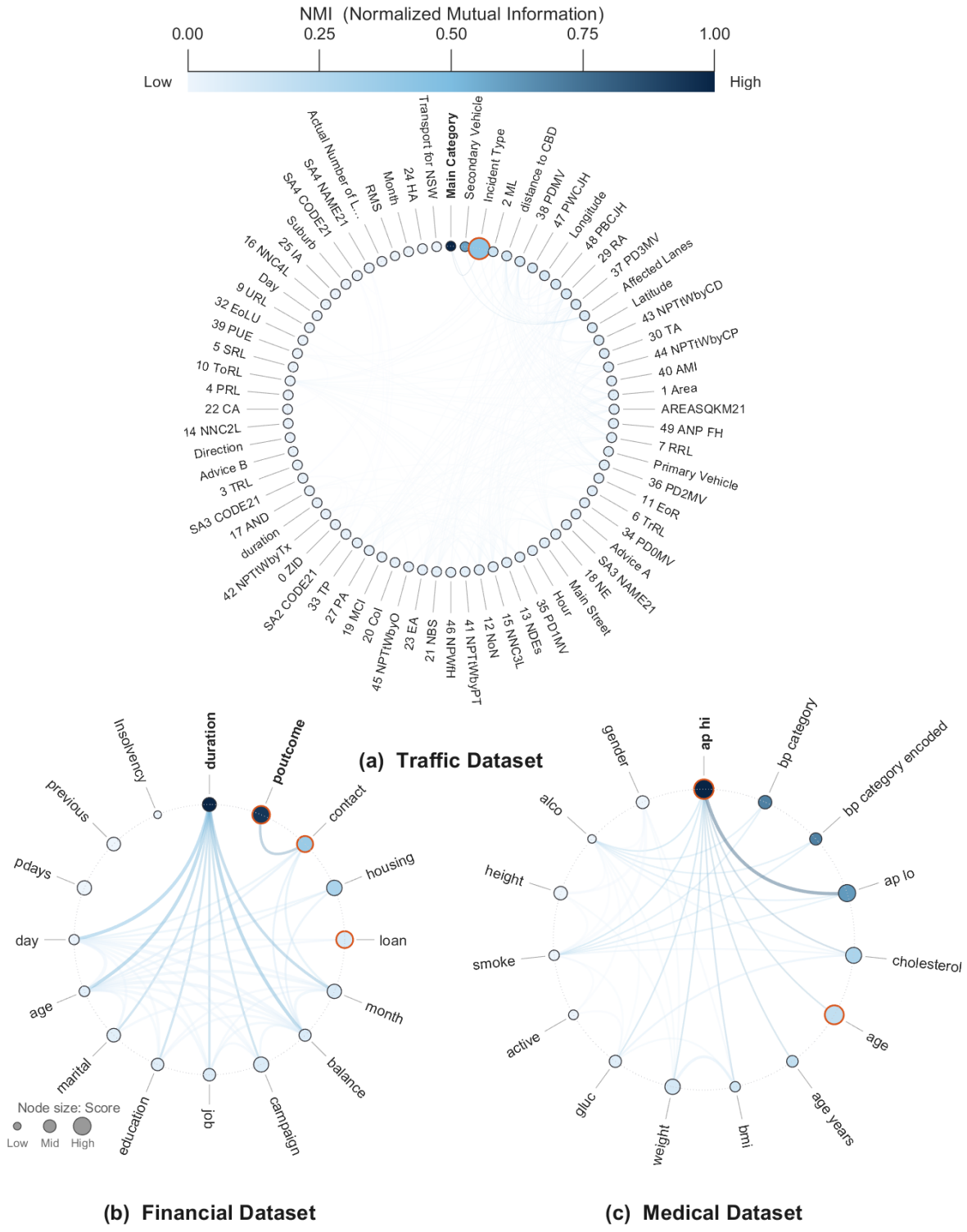


Fig. 6 Feature interaction networks derived from the CNFI module across three representative domains. Node size reflects Eigen-Rank centrality; fill color encodes NMI with the prediction target, ranging from light blue (low) to dark blue (high); orange-bordered nodes belong to the top 20 by composite score. The distinct topological structures across domains — from the hub-dominated financial network to the sparse traffic layout and densely clustered medical subgraph — visually corroborate the domain-specific nature of feature interactions. Subfigures (a)–(c) correspond to the Traffic, Financial and Medical domains, respectively.

the field's long-standing emphasis on ranking topology may have been misplaced relative to the more fundamental task of accurately quantifying pairwise feature interactions. The cross-domain comparison further validates the necessity of jointly modeling interaction structure alongside individual feature relevance, a design principle that our framework explicitly instantiates. This corroborates a foundational tenet of the feature selection literature: an optimal feature subset must simultaneously maximize target relevance, minimize inter-feature redundancy, and explicitly capture higher-order interaction effects [50, 51]. Methods that optimize for only a subset of these criteria inevitably exhibit domain-specific fragility. NMI [30], for instance, performs competitively on datasets whose predictive signal is largely attributable to independent feature contributions, yet degrades markedly in domains where interaction structure is the primary determinant of outcome, as is characteristic of financial datasets with complex co-movement patterns among variables. CNFI-NMI resolves this inherent tension through a principled rank-based fusion strategy that dynamically integrates both sources of information, enabling consistent and robust generalization across datasets with heterogeneous statistical characteristics—a property that neither purely relevance-driven nor purely interaction-driven baselines are able to achieve in isolation.

5.2 Computational Complexity Analysis

The computational viability of the CNFI-NMI framework is underpinned by the scalable nature of its two constituent modules. To be theoretically precise, the overall computational complexity can be systematically decomposed into three main components.

FP-Growth has complexity that is linear in the sample size N and the tree depth, roughly $O(N \cdot \text{TreeDepth})$ [44]. By contrast, iterative wrapper methods incur exponential or quadratic cost ($O(N^2)$ or higher) due to repeated model retraining. The discrete association-mining approach instead acts as an efficient one-pass filter that examines the interaction space without the prohibitive cost of candidate generation. In summary, FP-Growth keeps the interaction-mining stage at near-linear cost, whereas wrapper methods scale superlinearly.

6 Conclusion

Predicting outcomes from data with complex feature interactions is challenging, particularly in high-stakes domains such as finance and healthcare. Our findings show that combining network topology with information theory provides a robust solution: the proposed CNFI-NMI framework uses Composite Adaptive Weighting to map non-linear dependencies effectively without biasing relevance scores. CNFI-NMI outperforms current baselines across cross-domain tests, and its advantage is most pronounced in imbalanced data settings; at the same time, the framework provides important interpretability advantages across critical domains. The framework nonetheless has limitations. Future versions should explore network sampling or approximate association mining to address extremely large or complex datasets more effectively. Because discretization introduces information loss, kernel-based dependency measures could reduce this limitation significantly. Moreover, shifting the network toward causal discovery could eventually guide Graph Neural Network design. Overall, CNFI-NMI provides a reliable foundation for predictive modeling over complex tabular data.

Availability of Data and Materials

The data and source codes that support the findings of this study are openly available in GitHub at https://github.com/xux06814-cell/CNFI-NMI/tree/main.

Author Contributions

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Declaration of Competing Interest

The authors have no competing interests to declare that are relevant to the content of this article.

Electronic Supplementary Material

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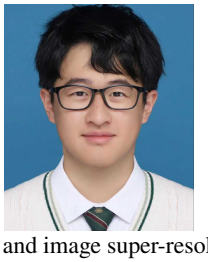
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